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Security Operations Center (SOC) Investigation Report

-Investigated by

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Investigation Title

Initial Network Traffic Triage and Protocol Analysis Using Wireshark

1. Executive Summary

A packet capture (PCAP) file named 2026-02-28-traffic-analysis-exercise.pcap was analyzed to develop practical SOC investigation skills and establish a structured workflow for network traffic analysis.

The investigation focused on identifying key hosts, understanding network protocols present in the environment, analyzing DNS activity, and reviewing HTTP communications to determine whether any obvious Indicators of compromise (IOCs) were present.

Analysis revealed a Windows Active Directory environment containing DNS, SMB, LDAP, Kerberos, HTTP, HTTPS, and Microsoft cloud-related communications. No immediately malicious domains, payloads, or command-and-control indicators were identified during the initial investigation phases.

The exercise emphasized methodology, traffic baselining, and investigative reasoning rather than confirming a security incident.

2. Investigation Objective

What We Had to Find

The objectives of this investigation were:

- Understand the network environment represented in the PCAP.
- Identify the primary communicating hosts.
- Determine the likely workstation and infrastructure systems.
- Review DNS activity for suspicious domains.
- Examine HTTP communications for unusual behavior.

- Develop a structured SOC triage workflow.
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3. Investigation Methodology

The following workflow was followed:

1. Protocol Hierarchy Analysis
 2. Conversation Analysis
 3. Endpoint Identification
 4. DNS Investigation
 5. HTTP Investigation
 6. Hypothesis Development
 7. Initial Threat Assessment
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4. Phase 1 – Protocol Hierarchy Analysis

Goal

To identify the major protocols present in the capture and understand the network environment.

Action Performed

Navigated to:

Statistics → Protocol Hierarchy

Findings

The following protocols were observed:

- TCP
- TLS
- SMB / SMB2
- LDAP
- Kerberos
- DNS
- HTTP

- QUIC
- NetBIOS

The presence of LDAP, Kerberos, SMB, and NetBIOS indicated a Windows Active Directory environment.

Result

The environment appeared to be a corporate Windows network utilizing Active Directory services.

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s
Frame	100.0	15512	100.0	6333775	3229	0	0	0
Ethernet	100.0	15512	4.1	259124	132	0	0	0
Internet Protocol Version 4	94.6	14676	4.6	293520	149	0	0	0
Transmission Control Protocol	80.9	12544	4.2	265596	135	7898	172568	87
Transport Layer Security	12.0	1857	46.0	2913926	1485	1857	1655854	844
NetBIOS Session Service	5.2	808	3.8	242449	123	0	0	0
SMB2 (Server Message Block Protocol version 2)	5.1	785	3.7	235652	120	721	206140	105
Distributed Computing Environment / Remote Procedure Call (DCE/RPC)	0.4	56	0.1	5696	2	10	1380	0
SAMR (pidl)	0.2	30	0.0	2204	1	30	2204	1
Local Security Authority	0.1	16	0.0	1008	0	12	336	0
Malformed Packet	0.0	4	0.0	0	0	4	0	0
SMB (Server Message Block Protocol)	0.1	23	0.1	3565	1	23	3565	1
Distributed Computing Environment / Remote Procedure Call (DCE/RPC)	4.9	760	3.6	231028	117	246	109932	56
Active Directory Replication	2.3	352	0.7	42496	21	352	42496	21
DCE/RPC Endpoint Mapper	0.9	132	0.4	25896	13	132	25896	13
Local Security Authority	0.2	24	0.1	3360	1	24	3360	1
Microsoft Network Logon	0.0	6	0.1	5520	2	6	5520	2
Lightweight Directory Access Protocol	3.3	519	5.7	362328	184	519	281940	143
Hypertext Transfer Protocol	2.3	354	18.3	1156329	589	59	22238	11
HTML Form URL Encoded	1.7	266	0.2	10001	5	266	10001	5
Media Type	0.2	25	16.8	1061903	541	25	1061903	541
Online Certificate Status Protocol	0.0	2	0.0	942	0	2	942	0
Line-based text data	0.0	2	0.0	103	0	2	103	0
JavaScript Object Notation	0.0	1	0.0	0	0	0	0	0
Data	2.2	340	0.1	8204	4	340	8204	4
Kerberos	0.1	16	0.4	22632	11	16	22632	11
User Datagram Protocol	13.6	2113	0.3	16904	8	0	0	0
Domain Name System	5.2	809	1.2	73858	37	809	73858	37
QUIC IETF	4.0	626	6.6	421009	214	626	419705	214
NetBIOS Datagram Service	2.8	430	0.6	35260	17	0	0	0
SMB (Server Message Block Protocol)	2.8	430	0.7	47498	24	0	0	0
SMB MailSlot Protocol	2.8	430	0.2	10750	5	0	0	0
Microsoft Windows Browser Protocol	2.8	430	0.2	10518	5	430	10518	5
NetBIOS Name Service	0.8	118	0.1	7718	3	118	7718	3
Connectionless Lightweight Directory Access Protocol	0.4	64	0.2	13417	6	64	13417	6
Network Time Protocol	0.2	28	0.1	3360	1	28	3360	1
Simple Service Discovery Protocol	0.2	27	0.1	3691	1	27	3691	1
Link-local Multicast Name Resolution	0.0	4	0.0	132	0	4	132	0
Dynamic Host Configuration Protocol	0.0	4	0.0	1281	0	4	1281	0
Multicast Domain Name System	0.0	3	0.0	120	0	3	120	0

5. Phase 2 – Conversation Analysis

Goal

To identify the most active hosts and determine which systems warranted further investigation.

Action Performed

Navigated to:

Statistics → Conversations → IPv4

Sorted by total bytes transferred.

Findings

Statistics → Endpoints → IPv4

Findings

Top communicating hosts included:

- 10.2.28.88
- 10.2.28.2
- Multiple Microsoft-owned IP addresses
- Multiple Akamai-owned IP addresses

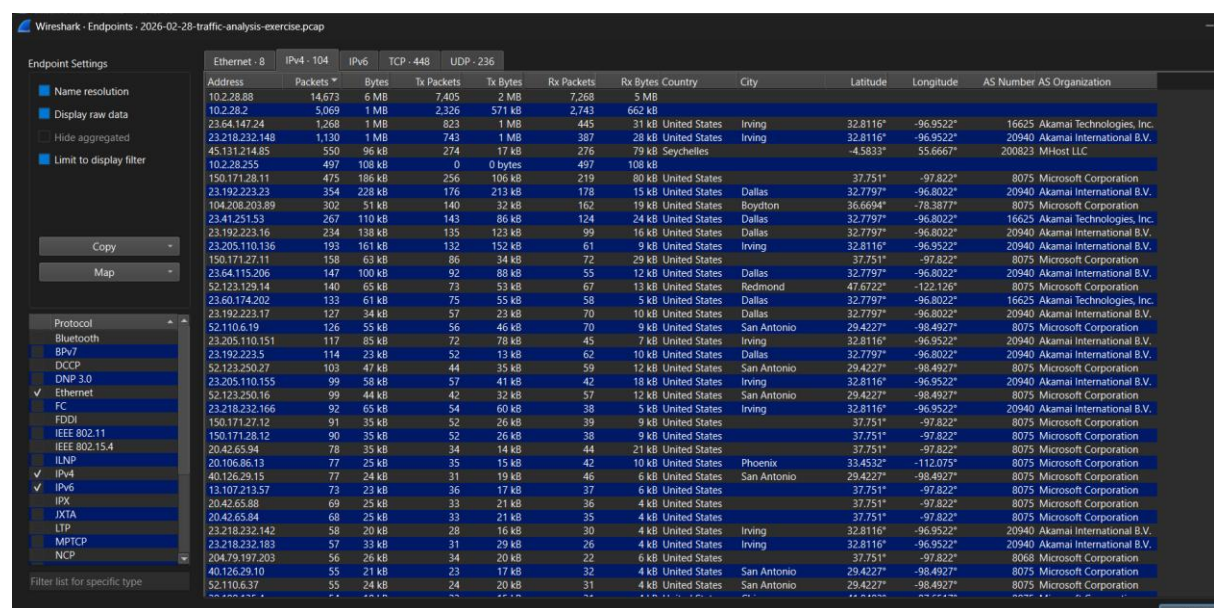
Analysis

The traffic pattern suggested:

- 10.2.28.88 = User Workstation
- 10.2.28.2 = Internal DNS and potentially Active Directory infrastructure

Result

The network environment was mapped sufficiently for protocol-specific investigation.



Wireshark - Endpoints - 2026-02-28-traffic-analysis-exercise.pcap

Endpoint Settings

- ☒ Name resolution
- ☒ Display raw data
- ☐ Hide aggregated
- ☒ Limit to display filter

Copy

Map

Protocol

- ☐ Bluetooth
- ☐ BPV7
- ☐ DCCP
- ☐ DNP 3.0
- ☒ Ethernet
- ☐ FC
- ☐ FDDI
- ☐ IEEE 802.11
- ☐ IEEE 802.15.4
- ☐ ILNP
- ☒ IPv4
- ☒ IPv6
- ☐ JXTA
- ☐ LTP
- ☐ MPTCP
- ☐ NCP

Filter list for specific type

Address	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country	City	Latitude	Longitude	AS Number	AS Organization
10.2.28.88	14,673	6 MB	7,405	2 MB	7,268	5 MB						
10.2.28.2	5,069	1 MB	2,326	571 kB	2,743	662 kB						
23.64.147.24	1,268	1 MB	823	1 MB	445	31 kB	United States	Irving	32.8116°	-96.9522°	16625	Akamai Technologies, Inc.
23.218.232.148	1,130	1 MB	743	1 MB	387	28 kB	United States	Irving	32.8116°	-96.9522°	20940	Akamai International B.V.
45.131.214.85	550	96 kB	274	17 kB	276	79 kB	Seychelles		-4.5833°	55.6667°	200823	MiHost LLC
10.2.28.255	497	108 kB	0	0 bytes	497	108 kB						
150.171.28.11	475	186 kB	256	106 kB	219	80 kB	United States		37.751°	-97.822°	8075	Microsoft Corporation
23.192.233.23	354	228 kB	176	213 kB	178	15 kB	United States	Dallas	32.7797°	-96.8022°	20940	Akamai International B.V.
104.208.203.89	302	51 kB	140	32 kB	162	19 kB	United States	Boydton	36.6694°	-78.3877°	8075	Microsoft Corporation
23.41.251.53	267	110 kB	143	86 kB	124	24 kB	United States	Dallas	32.7797°	-96.8022°	16625	Akamai Technologies, Inc.
23.192.233.16	234	138 kB	135	123 kB	99	16 kB	United States	Dallas	32.7797°	-96.8022°	20940	Akamai International B.V.
23.205.110.136	193	161 kB	132	152 kB	61	9 kB	United States	Irving	32.8116°	-96.9522°	20940	Akamai International B.V.
150.171.27.11	158	63 kB	86	34 kB	72	29 kB	United States		37.751°	-97.822°	8075	Microsoft Corporation
23.64.115.206	147	100 kB	92	80 kB	55	12 kB	United States	Dallas	32.7797°	-96.8022°	20940	Akamai International B.V.
52.123.129.14	140	65 kB	73	53 kB	67	13 kB	United States	Redmond	47.6722°	-122.126°	8075	Microsoft Corporation
23.60.174.202	133	61 kB	75	55 kB	58	5 kB	United States	Dallas	32.7797°	-96.8022°	16625	Akamai Technologies, Inc.
23.192.233.17	127	34 kB	57	23 kB	70	10 kB	United States	Dallas	32.7797°	-96.8022°	20940	Akamai International B.V.
52.110.6.19	126	55 kB	56	46 kB	70	9 kB	United States	San Antonio	29.4227°	-98.4927°	8075	Microsoft Corporation
23.205.110.151	117	85 kB	72	78 kB	45	7 kB	United States	Irving	32.8116°	-96.9522°	20940	Akamai International B.V.
23.192.233.5	114	23 kB	52	13 kB	62	10 kB	United States	Dallas	32.7797°	-96.8022°	20940	Akamai International B.V.
52.123.250.27	103	47 kB	44	35 kB	59	12 kB	United States	San Antonio	29.4227°	-98.4927°	8075	Microsoft Corporation
23.205.110.155	99	58 kB	57	41 kB	42	18 kB	United States	Irving	32.8116°	-96.9522°	20940	Akamai International B.V.
52.123.250.16	99	44 kB	42	32 kB	57	12 kB	United States	San Antonio	29.4227°	-98.4927°	8075	Microsoft Corporation
23.128.232.166	92	65 kB	54	60 kB	38	5 kB	United States	Irving	32.8116°	-96.9522°	20940	Akamai International B.V.
150.171.27.12	91	35 kB	52	26 kB	39	9 kB	United States		37.751°	-97.822°	8075	Microsoft Corporation
150.171.28.12	90	35 kB	52	26 kB	38	9 kB	United States		37.751°	-97.822°	8075	Microsoft Corporation
20.42.65.94	78	35 kB	34	14 kB	44	21 kB	United States		37.751°	-97.822°	8075	Microsoft Corporation
20.106.86.13	77	25 kB	35	15 kB	42	10 kB	United States	Phoenix	33.4532°	-112.075°	8075	Microsoft Corporation
40.126.29.15	77	24 kB	31	19 kB	46	6 kB	United States	San Antonio	29.4227°	-98.4927°	8075	Microsoft Corporation
13.107.213.57	73	23 kB	36	17 kB	37	6 kB	United States		37.751°	-97.822°	8075	Microsoft Corporation
20.42.65.88	69	25 kB	33	21 kB	36	4 kB	United States		37.751°	-97.822°	8075	Microsoft Corporation
20.42.65.84	68	25 kB	33	21 kB	35	4 kB	United States		37.751°	-97.822°	8075	Microsoft Corporation
23.218.232.142	59	20 kB	28	16 kB	30	4 kB	United States	Irving	32.8116°	-96.9522°	20940	Akamai International B.V.
23.218.232.183	57	33 kB	31	29 kB	26	4 kB	United States	Irving	32.8116°	-96.9522°	20940	Akamai International B.V.
204.79.197.203	56	26 kB	34	20 kB	22	6 kB	United States		37.751°	-97.822°	8068	Microsoft Corporation
40.126.29.10	55	21 kB	23	17 kB	32	4 kB	United States	San Antonio	29.4227°	-98.4927°	8075	Microsoft Corporation
52.110.6.37	55	24 kB	24	20 kB	31	4 kB	United States	San Antonio	29.4227°	-98.4927°	8075	Microsoft Corporation

7. Phase 4 – DNS Investigation

Goal

To identify suspicious domains, beaconing activity, or DNS tunneling behavior.

Action Performed

Applied filter:

dns

Reviewed DNS conversations and responses.

Findings

DNS responses included:

- login.microsoftonline.com
- officeclient.microsoft.com
- odc.officeapps.live.com
- www.bing.com
- Microsoft Edge-related services

No obviously malicious or randomly generated domains were identified.

Analysis

The observed DNS traffic appeared consistent with normal Microsoft Windows and Office 365 activity.

Result

No evidence of suspicious DNS activity or DNS-based command-and-control communications was identified.

No.	Time	Source	Destination	Protocol	Length	Info
7	2.473183	10.2.28.88	easyas123-dc.easysas...	DNS	122	Standard query 0x7c4c SRV _ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.mshome.net
8	2.473185	10.2.28.88	easyas123-dc.easysas...	DNS	126	Standard query response 0x7c4c SRV _ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.mshome.net
9	2.473186	10.2.28.88	easyas123-dc.easysas...	DNS	116	Standard query 0x62f5 SRV _ldap._tcp.Default-First-Site-Name._sites.easysas123.tech
10	2.473483	easyas123-dc.easysas...	10.2.28.88	DNS	189	Standard query response 0x230d SRV _ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.easysas123.tech SRV 0 100 389 easyas123-dc.easysas123.tech
11	2.473544	easyas123-dc.easysas...	10.2.28.88	DNS	179	Standard query response 0x62f5 SRV _ldap._tcp.Default-First-Site-Name._sites.easysas123.tech SRV 0 100 389 easyas123-dc.easysas123.tech
12	2.473589	10.2.28.88	easyas123-dc.easysas...	DNS	87	Standard query 0xbabd A easyas123-dc.easysas123.tech
13	2.473890	easyas123-dc.easysas...	10.2.28.88	DNS	103	Standard query response 0xbabd A easyas123-dc.easysas123.tech A 10.2.28.2
18	2.507994	10.2.28.88	easyas123-dc.easysas...	DNS	122	Standard query 0x7c4c SRV _ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.mshome.net
19	2.504461	10.2.28.88	easyas123-dc.easysas...	DNS	126	Standard query 0xb5fd SRV _ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.easysas123.tech
20	2.524463	10.2.28.88	easyas123-dc.easysas...	DNS	116	Standard query 0xb5fd SRV _ldap._tcp.Default-First-Site-Name._sites.easysas123.tech
21	2.524662	easyas123-dc.easysas...	10.2.28.88	DNS	189	Standard query response 0xb5fd SRV _ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.easysas123.tech SRV 0 100 389 easyas123-dc.easysas123.tech
22	2.524664	easyas123-dc.easysas...	10.2.28.88	DNS	179	Standard query response 0xb5fd SRV _ldap._tcp.Default-First-Site-Name._sites.easysas123.tech SRV 0 100 389 easyas123-dc.easysas123.tech
27	2.536640	easyas123-dc.easysas...	10.2.28.88	DNS	208	Standard query response 0x7c4c No such name SRV _ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.mshome.net SOA ns1-35.azure-dns.com
28	2.536642	10.2.28.88	easyas123-dc.easysas...	DNS	91	Standard query 0x1f60 SRV _ldap._tcp.dc._msdcs.mshome.net
29	2.576406	10.2.28.88	easyas123-dc.easysas...	DNS	91	Standard query 0x1f60 SRV _ldap._tcp.dc._msdcs.mshome.net
41	2.597177	easyas123-dc.easysas...	10.2.28.88	DNS	177	Standard query response 0x1f60 No such name SRV _ldap._tcp.dc._msdcs.mshome.net SOA ns1-35.azure-dns.com
42	2.672359	10.2.28.88	easyas123-dc.easysas...	DNS	79	Standard query 0xc491 A wpad.easysas123.tech
43	2.672361	10.2.28.88	easyas123-dc.easysas...	DNS	79	Standard query 0xc340 A wpad.easysas123.tech
44	2.672557	easyas123-dc.easysas...	10.2.28.88	DNS	153	Standard query response 0xc340 No such name A wpad.easysas123.tech SOA easyas123-dc.easysas123.tech
45	2.672558	easyas123-dc.easysas...	10.2.28.88	DNS	153	Standard query response 0xc491 No such name A wpad.easysas123.tech SOA easyas123-dc.easysas123.tech
46	2.672847	10.2.28.88	easyas123-dc.easysas...	DNS	75	Standard query 0x80bf A wpad.mshome.net
47	2.672849	10.2.28.88	easyas123-dc.easysas...	DNS	75	Standard query 0x5528 A wpad.mshome.net
48	2.726544	easyas123-dc.easysas...	10.2.28.88	DNS	161	Standard query response 0x80bf No such name A wpad.mshome.net SOA ns1-35.azure-dns.com
49	2.726545	easyas123-dc.easysas...	10.2.28.88	DNS	161	Standard query response 0x5528 No such name A wpad.mshome.net SOA ns1-35.azure-dns.com
57	3.103990	10.2.28.88	easyas123-dc.easysas...	DNS	122	Standard query 0x6900 SRV _ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.mshome.net
58	3.104318	easyas123-dc.easysas...	10.2.28.88	DNS	208	Standard query response 0x6900 No such name SRV _ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.mshome.net SOA ns1-35.azure-dns.com
59	3.104353	10.2.28.88	easyas123-dc.easysas...	DNS	91	Standard query 0x53ee SRV _ldap._tcp.dc._msdcs.mshome.net
60	3.104597	easyas123-dc.easysas...	10.2.28.88	DNS	177	Standard query response 0x53ee No such name SRV _ldap._tcp.dc._msdcs.mshome.net SOA ns1-35.azure-dns.com
81	3.222279	10.2.28.88	easyas123-dc.easysas...	DNS	83	Standard query 0x3113 A www.msftconnecttest.com
82	3.249684	10.2.28.88	easyas123-dc.easysas...	DNS	83	Standard query 0x3113 A www.msftconnecttest.com
83	3.262593	easyas123-dc.easysas...	10.2.28.88	DNS	227	Standard query response 0x3113 A www.msftconnecttest.com CNAME ncsi-geo.trafficmanager.net CNAME www.msftncsi.com.edgesuite.net

8. Phase 5 – HTTP Investigation

Goal

To identify potential file downloads, suspicious web requests, or unusual HTTP activity.

Action Performed

Applied filter:

http

Reviewed HTTP conversations.

Findings

Several HTTP conversations were identified.

One external IP address, 45.131.214.85, was selected for further investigation due to its appearance outside the previously observed Microsoft and Akamai infrastructure.

Analysis

The IP was classified as an investigative lead rather than a confirmed malicious indicator.

No malicious payloads or confirmed indicators of compromise were identified during the initial review.

Result

Further stream analysis was recommended for complete validation.

No.	http	e	Source	Destination	Protocol	Length	Info
189329	http3		10.2.28.88	a1961.g2.akamai.net	HTTP	208	GET /connecttest.txt HTTP/1.1
2638	45.131.214.85		10.2.28.88	vadusa.xyz	HTTP	241	HTTP/1.1 200 OK (text/plain)
2639	45.195238		10.2.28.88	vadusa.xyz	HTTP	272	POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
2640	45.196891		10.2.28.88	vadusa.xyz	HTTP	278	HTTP/1.1 200 OK (application/x-www-form-urlencoded)
2641	45.355474		10.2.28.88	vadusa.xyz	HTTP	494	POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
2643	45.555924		10.2.28.88	vadusa.xyz	HTTP	362	HTTP/1.1 200 OK (application/x-www-form-urlencoded)
2645	45.756275		10.2.28.88	vadusa.xyz	HTTP	326	POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
3807	105.932070		10.2.28.88	vadusa.xyz	HTTP	327	POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
4093	166.005869		10.2.28.88	vadusa.xyz	HTTP	286	POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
4134	226.086899		10.2.28.88	vadusa.xyz	HTTP	286	POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
4149	286.161920		10.2.28.88	vadusa.xyz	HTTP	286	POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
4201	346.346182		10.2.28.88	vadusa.xyz	HTTP	286	POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
4211	406.426196		10.2.28.88	vadusa.xyz	HTTP	286	POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
4215	466.505973		10.2.28.88	vadusa.xyz	HTTP	286	POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
4225	526.688782		10.2.28.88	vadusa.xyz	HTTP	286	POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
4227	586.765212		10.2.28.88	vadusa.xyz	HTTP	286	POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
4270	646.858958		10.2.28.88	vadusa.xyz	HTTP	286	POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
4286	707.044699		10.2.28.88	vadusa.xyz	HTTP	286	POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
4299	737.770919		10.2.28.88	a122.dscd.akamai.net	HTTP	412	GET /assets/Owner/arm/ProcessMAU.txt HTTP/1.1
4301	737.809659		10.2.28.88	a122.dscd.akamai.net	HTTP	342	HTTP/1.1 304 Not Modified
4448	767.221284		10.2.28.88	vadusa.xyz	HTTP	286	POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
4450	827.402831		10.2.28.88	vadusa.xyz	HTTP	286	POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
4470	887.686814		10.2.28.88	vadusa.xyz	HTTP	286	POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
4638	947.874128		10.2.28.88	vadusa.xyz	HTTP	286	POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
4737	1008.057273		10.2.28.88	vadusa.xyz	HTTP	286	POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
4770	1068.138676		10.2.28.88	vadusa.xyz	HTTP	286	POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
4785	1128.228023		10.2.28.88	vadusa.xyz	HTTP	286	POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
4812	1188.308378		10.2.28.88	vadusa.xyz	HTTP	286	POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
4820	1248.491043		10.2.28.88	vadusa.xyz	HTTP	286	POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
4952	1308.571283		10.2.28.88	vadusa.xyz	HTTP	286	POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)

Frame 87: Packet, 208 bytes on wire (1664 bits), 208 bytes captured (1664 bits)

Ethernet II, Src: 00:19:d1:b2:4d:ad, Dst: 00:05:00:41:ae:25

Internet Protocol Version 4, Src: 10.2.28.88 (10.2.28.88), Dst: a1961.g2.akamai.net (23.47.50.182)

Transmission Control Protocol, Src Port: 59982, Dst Port: 80, Seq: 1, Ack: 1, Len: 154

Hypertext Transfer Protocol

9. Key Learning Outcomes

During this investigation the following SOC concepts were practiced:

- Protocol hierarchy interpretation
 - Endpoint identification
 - DNS analysis methodology
 - HTTP traffic triage
 - Hypothesis-driven investigation
 - Differentiating normal and suspicious behavior
 - Building investigative reasoning
-

10. Final Assessment

Observed

- Windows Active Directory traffic
- Microsoft cloud communications
- DNS activity
- HTTP activity
- SMB and LDAP communications

Not Observed

- Malware downloads
 - Command-and-control traffic
 - DNS tunneling
 - Data exfiltration
 - Confirmed malicious domains
 - Confirmed indicators of compromise
-

11. Conclusion

This investigation focused on establishing a structured SOC workflow for packet analysis rather than identifying a confirmed security incident.

The exercise demonstrated the importance of understanding normal network behavior before attempting to identify malicious activity. Through protocol analysis, endpoint identification, DNS review, and HTTP investigation, a baseline understanding of the environment was established.

The investigation highlighted a critical SOC principle:

Understanding normal traffic patterns is the foundation for detecting abnormal and potentially malicious activity.